AQA A-Level Further Mathematics (7367) — Topic 1.5

1.5: Further Calculus

1. Arc Length

Arc length (Cartesian): The length of the curve $y = f(x)$ from $x = a$ to $x = b$:

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

Arc length (parametric): If $x = x(t)$, $y = y(t)$, from $t = t_1$ to $t = t_2$:

$$L = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

Example: Find the arc length of $y = \frac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$.

$\frac{dy}{dx} = x^{1/2}$, so $1 + \left(\frac{dy}{dx}\right)^2 = 1 + x$

$L = \int_0^3 \sqrt{1+x}\, dx = \left[\frac{2}{3}(1+x)^{3/2}\right]_0^3 = \frac{2}{3}(4^{3/2}) - \frac{2}{3}(1) = \frac{2}{3}(8) - \frac{2}{3} = \frac{14}{3}$

AQA Exam Tip: AQA often chooses curves where $1 + (dy/dx)^2$ simplifies to a perfect square. If your integrand doesn't simplify nicely, recheck your differentiation.

2. Surface Area of Revolution

Surface area of revolution about the x-axis:

$$S = 2\pi \int_a^b y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

Parametric form:

$$S = 2\pi \int_{t_1}^{t_2} y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

Example: Find the surface area generated when the curve $y = x^2$ from $x = 0$ to $x = 1$ is rotated 2π about the x-axis.

$\frac{dy}{dx} = 2x$, so $1 + (2x)^2 = 1 + 4x^2$

$S = 2\pi \int_0^1 x^2\sqrt{1 + 4x^2}\, dx$
(This integral requires substitution $x = \frac{1}{2}\tan\theta$ — see integration section for technique.)

Result after full working: $S = \frac{\pi}{6}\left(5\sqrt{5} - 1\right)$

3. Mean Value of a Function

Mean value: The mean value of $f(x)$ over $[a, b]$ is:

$$\bar{f} = \frac{1}{b - a}\int_a^b f(x)\, dx$$

Example: Find the mean value of $f(x) = x^2 + 1$ over $[0, 3]$.

$$\bar{f} = \frac{1}{3}\int_0^3 (x^2+1)\,dx = \frac{1}{3}\left[\frac{x^3}{3}+x\right]_0^3 = \frac{1}{3}(9+3) = 4$$

4. Reduction Formulae

A reduction formula expresses an integral I_n in terms of I_{n-1} (or I_{n-2}), allowing evaluation by repeatedly reducing the index.

Example: Derive a reduction formula for $I_n = \displaystyle\int_0^{\pi/2} \sin^n x\,dx$.

Write $\sin^n x = \sin^{n-1} x \cdot \sin x$ and integrate by parts:
Let $u = \sin^{n-1} x$, $dv = \sin x\,dx \Rightarrow v = -\cos x$

$$I_n = \left[-\cos x\,\sin^{n-1} x\right]_0^{\pi/2} + (n-1)\int_0^{\pi/2} \cos^2 x\,\sin^{n-2} x\,dx$$
$$= 0 + (n-1)\int_0^{\pi/2}(1-\sin^2 x)\sin^{n-2} x\,dx = (n-1)(I_{n-2} - I_n)$$

$$I_n + (n-1)I_n = (n-1)I_{n-2} \Rightarrow \boxed{I_n = \frac{n-1}{n} I_{n-2}}$$

With: $I_0 = \dfrac{\pi}{2}$, $I_1 = 1$.

So: $I_2 = \dfrac{1}{2}\cdot\dfrac{\pi}{2} = \dfrac{\pi}{4}$, $I_4 = \dfrac{3}{4}\cdot\dfrac{\pi}{4} = \dfrac{3\pi}{16}$, $I_3 = \dfrac{2}{3}$, $I_5 = \dfrac{8}{15}$, etc.

Example: Derive a reduction formula for $I_n = \displaystyle\int x^n e^x\,dx$.

Integrate by parts: $u = x^n$, $dv = e^x dx$
$I_n = x^n e^x - n\int x^{n-1} e^x\,dx = x^n e^x - n\,I_{n-1}$

Reduction formula: $I_n = x^n e^x - n\,I_{n-1}$
Starting from $I_0 = e^x + C$.

AQA Exam Tip: When asked to "prove" a reduction formula, show integration by parts in full. When asked to "use" it, apply it repeatedly, citing the formula each time. AQA often asks for a specific value like I_4 or I_5 — make sure you track the base cases.

5. Improper Integrals

An integral is **improper** if:

- The interval of integration is infinite: $\int_a^\infty f(x)\,dx = \lim_{t\to\infty}\int_a^t f(x)\,dx$
- The integrand has a discontinuity in $[a, b]$: $\int_a^b f(x)\,dx = \lim_{\epsilon\to 0^+}\int_{a+\epsilon}^b f(x)\,dx$

The integral **converges** if the limit exists and is finite; otherwise it **diverges**.

Example: Determine whether $\int_1^\infty \frac{1}{x^2}\,dx$ converges.

$\int_1^t x^{-2}\,dx = \left[-x^{-1}\right]_1^t = -\frac{1}{t} + 1$

As $t \to \infty$: $1 - \frac{1}{t} \to 1$. Converges to 1.

Example: Show that $\int_1^\infty \frac{1}{x}\,dx$ diverges.

$\int_1^t \frac{1}{x}\,dx = [\ln x]_1^t = \ln t \to \infty$ as $t \to \infty$. Diverges.

p-test: $\int_1^\infty x^{-p}\,dx$ converges if and only if $p > 1$. Similarly, $\int_0^1 x^{-p}\,dx$ converges if and only if $p < 1$.

6. Integration Techniques

Integration by Partial Fractions

Decompose the rational function into partial fractions, then integrate each term.

Example: Find $\int \frac{3}{(x-1)(x+2)}\,dx$.

$$\frac{3}{(x-1)(x+2)} = \frac{1}{x-1} - \frac{1}{x+2}$$

$$\int = \ln|x-1| - \ln|x+2| + C = \ln\left|\frac{x-1}{x+2}\right| + C$$

Inverse Trigonometric Integrals

$$\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \arcsin\left(\frac{x}{a}\right) + C, \quad |x| < a$$

$$\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$$

Example: Find $\int \frac{1}{\sqrt{9-x^2}}\,dx$.

Using the formula with $a = 3$: $\arcsin\left(\frac{x}{3}\right) + C$

Example: Find $\int_0^2 \frac{1}{4+x^2}\,dx$.

$$= \left[\frac{1}{2}\arctan\left(\frac{x}{2}\right)\right]_0^2 = \frac{1}{2}\arctan(1) - 0 = \frac{1}{2}\cdot\frac{\pi}{4} = \frac{\pi}{8}$$

Completing the Square for Integration

Integrals of the form $\dfrac{1}{ax^2+bx+c}$ or $\dfrac{1}{\sqrt{ax^2+bx+c}}$ require completing the square to match standard forms.

Example: Find $\displaystyle\int \frac{1}{x^2+6x+13}\,dx$.

Complete the square: $x^2+6x+13=(x+3)^2+4$

$$\int \frac{1}{(x+3)^2+4}\,dx = \frac{1}{2}\arctan\left(\frac{x+3}{2}\right)+C$$

7. Summary of Key Results

Result	Formula
Arc length (Cartesian)	$\displaystyle\int \sqrt{1+(dy/dx)^2}\,dx$
Arc length (parametric)	$\displaystyle\int \sqrt{(dx/dt)^2+(dy/dt)^2}\,dt$
Surface area (x-axis)	$\displaystyle 2\pi\int y\sqrt{1+(dy/dx)^2}\,dx$
Mean value	$\displaystyle\frac{1}{b-a}\int_a^b f(x)\,dx$
$\int 1/\sqrt{a^2-x^2}\,dx$	$\arcsin(x/a)+C$
$\int 1/(a^2+x^2)\,dx$	$\frac{1}{a}\arctan(x/a)+C$
Wallis reduction	$I_n=\dfrac{n-1}{n}I_{n-2}$ for $\int_0^{\pi/2}\sin^n x\,dx$

8. Practice Checklist

- Can you calculate arc length and surface area for both Cartesian and parametric curves?
- Can you find the mean value of a function over an interval?
- Can you derive and apply reduction formulae for $\int \sin^n x\,dx$, $\int \cos^n x\,dx$, and $\int x^n e^x\,dx$?
- Can you evaluate improper integrals and determine convergence/divergence?
- Can you integrate using inverse trigonometric functions and completing the square?

Arc Length — Parametric Examples

For a parametric curve $x = f(t)$, $y = g(t)$, the arc length from $t = a$ to $t = b$ is:

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$$

Example: Find the arc length of $x = t^3/3$, $y = t^2/2$ from $t = 0$ to $t = 2$. $\frac{dx}{dt} = t^2$, $\frac{dy}{dt} = t$. Then:

$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = t^4 + t^2 = t^2(t^2+1)$$

$$L = \int_0^2 t\sqrt{t^2+1}\,dt$$

Let $u = t^2 + 1$, $du = 2t\,dt$:

$$L = \frac{1}{2}\int_1^5 \sqrt{u}\,du = \frac{1}{2}\cdot\frac{2}{3}\left[u^{3/2}\right]_1^5 = \frac{1}{3}(5\sqrt{5} - 1)$$

Cartesian example: For $y = \cosh x$, $0 \le x \le 1$: $\frac{dy}{dx} = \sinh x$, and $1 + \sinh^2 x = \cosh^2 x$, so:

$$L = \int_0^1 \cosh x\,dx = \sinh 1$$

Surface of Revolution — Full Example

When $y = f(x)$ is rotated about the x-axis from $x = a$ to $x = b$:

$$S = 2\pi \int_a^b y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

Example: Find the surface area when $y = \sqrt{x}$, $1 \le x \le 4$, is rotated about the x-axis. $\frac{dy}{dx} = \frac{1}{2\sqrt{x}}$, so $1 + \left(\frac{dy}{dx}\right)^2 = \frac{4x+1}{4x}$.

$$S = 2\pi \int_1^4 \sqrt{x} \cdot \frac{\sqrt{4x+1}}{2\sqrt{x}}\, dx = \pi \int_1^4 \sqrt{4x+1}\, dx$$

Let $u = 4x + 1$: $S = \frac{\pi}{4} \cdot \frac{2}{3}\left[u^{3/2}\right]_5^{17} = \frac{\pi}{6}(17\sqrt{17} - 5\sqrt{5})$.

Reduction Formulae and Wallis's Formula

Let $I_n = \int_0^{\pi/2} \sin^n x\, dx$. The reduction formula (derived by IBP) is:

$$I_n = \frac{n-1}{n} I_{n-2}, \qquad I_0 = \frac{\pi}{2}, \quad I_1 = 1$$

Applying repeatedly gives **Wallis's formula**:

$$\frac{\pi}{2} = \frac{2 \cdot 2 \cdot 4 \cdot 4 \cdot 6 \cdot 6 \cdots}{1 \cdot 3 \cdot 3 \cdot 5 \cdot 5 \cdot 7 \cdots}$$

Values:

$$I_6 = \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{5\pi}{32}, \qquad I_5 = \frac{4}{5} \cdot \frac{2}{3} \cdot 1 = \frac{8}{15}$$

Integration Using Hyperbolic Substitution

Integrand form	Substitution	Result

$\dfrac{1}{\sqrt{a^2+x^2}}$	$x = a\sinh t$	$\operatorname{arcsinh}(x/a) + C = \ln\left(\frac{x}{a} + \sqrt{1+\frac{x^2}{a^2}}\right) + C$
$\dfrac{1}{\sqrt{x^2-a^2}}$	$x = a\cosh t$	$\operatorname{arccosh}(x/a) + C = \ln\left(x + \sqrt{x^2-a^2}\right) + C$
$\sqrt{a^2+x^2}$	$x = a\sinh t$	$\dfrac{x\sqrt{a^2+x^2}}{2} + \dfrac{a^2}{2}\operatorname{arcsinh}(x/a) + C$

Example: Find $\displaystyle\int_0^1 \frac{1}{\sqrt{x^2+3}}\,dx$. Let $x = \sqrt{3}\sinh t$: result is $\left[\operatorname{arcsinh}(x/\sqrt{3})\right]_0^1 = \operatorname{arcsinh}(1/\sqrt{3}) = \ln(1/\sqrt{3} + \sqrt{4/3}) = \ln(\sqrt{3}/\sqrt{3}\cdot\sqrt{3}) = \ln\sqrt{3}$.

AQA Exam Technique — Further Calculus

Key tips:

- **Arc length:** Remember the $\sqrt{}$ around $1 + (y')^2$. Simplify inside before integrating.
- **Surface area:** Has 2π and y (radius), not y^2. Check you are rotating about the correct axis.
- **Reduction formulae:** Derive by IBP unless told to assume. Show boundary terms vanish.
- **Improper integrals:** Write as a limit explicitly: $\displaystyle\lim_{t\to\infty}\int_a^t f(x)\,dx$.
- **Convergence:** $\int_1^\infty x^{-p}\,dx$ converges iff $p > 1$.
- **Mean value:** Mean $= \dfrac{1}{b-a}\int_a^b f(x)\,dx$. Don't omit the $\frac{1}{b-a}$ factor.

Volume of Revolution — Full Examples

Volume about the x-axis: $V = \pi\displaystyle\int_a^b y^2\,dx$

Volume about the y-axis: $V = \pi\displaystyle\int_c^d x^2\,dy$

Parametric form (x-axis): $V = \pi \int_{t_1}^{t_2} y^2 \frac{dx}{dt}\, dt$

Example 1: Find the volume generated when the region bounded by $y = \sqrt{x}$, $x = 0$, $x = 4$ is rotated 2π about the x-axis.

$$V = \pi \int_0^4 (\sqrt{x})^2\, dx = \pi \int_0^4 x\, dx = \pi \left[\frac{x^2}{2}\right]_0^4 = \pi \cdot 8 = 8\pi$$

Example 2: The curve $x = t^2,\ y = t^3$ for $0 \le t \le 1$ is rotated about the x-axis. Find the volume.

$\frac{dx}{dt} = 2t$, so $V = \pi \int_0^1 (t^3)^2 \cdot 2t\, dt = 2\pi \int_0^1 t^7\, dt = 2\pi \left[\frac{t^8}{8}\right]_0^1 = \frac{\pi}{4}$

Example 3: The region bounded by $y = e^x$, $y = 0$, $x = 0$, $x = 1$ is rotated about the x-axis. Find the volume.

$$V = \pi \int_0^1 e^{2x}\, dx = \pi \left[\frac{e^{2x}}{2}\right]_0^1 = \frac{\pi}{2}(e^2 - 1)$$

Integration by Parts — Advanced Techniques

Integration by parts: $\int u \frac{dv}{dx}\, dx = uv - \int v \frac{du}{dx}\, dx$

Use the **LIATE** priority for u: Logarithm, Inverse trig, Algebraic, Trigonometric, Exponential.

Example 1: $\int x^2 e^x\, dx$

Apply IBP twice. Let $u = x^2,\ dv = e^x\, dx$. Then $\int x^2 e^x\, dx = x^2 e^x - 2\int x e^x\, dx$.

For $\int x e^x\, dx$: $u = x,\ dv = e^x\, dx \Rightarrow x e^x - e^x$.

So $\int x^2 e^x\, dx = x^2 e^x - 2(x e^x - e^x) + C = e^x(x^2 - 2x + 2) + C$.

Example 2: $\int e^x \sin x\, dx$ (cyclic IBP)

Let $I = \int e^x \sin x\, dx$. IBP: $u = \sin x,\ dv = e^x\, dx$:

$I = e^x \sin x - \int e^x \cos x\, dx$.

IBP again on $\int e^x \cos x\,dx$: $u = \cos x,\ dv = e^x\,dx$:
$\int e^x \cos x\,dx = e^x \cos x + \int e^x \sin x\,dx = e^x \cos x + I$.
So $I = e^x \sin x - e^x \cos x - I \Rightarrow 2I = e^x(\sin x - \cos x) \Rightarrow I = \dfrac{e^x(\sin x - \cos x)}{2} + C$.

Example 3: $\int \ln x\,dx$
Let $u = \ln x,\ dv = 1\,dx \Rightarrow du = \frac{1}{x}\,dx,\ v = x$.
$\int \ln x\,dx = x\ln x - \int x \cdot \frac{1}{x}\,dx = x\ln x - x + C$.

Improper Integrals — Full Worked Examples

An integral is **improper** if:

- One or both limits are infinite: $\displaystyle\int_a^\infty f(x)\,dx = \lim_{t\to\infty}\int_a^t f(x)\,dx$
- The integrand has a singularity in $[a, b]$: $\displaystyle\int_0^1 x^{-1/2}\,dx = \lim_{\varepsilon\to 0^+}\int_\varepsilon^1 x^{-1/2}\,dx$

The integral **converges** if the limit exists and is finite; otherwise it **diverges**.

Example 1: $\displaystyle\int_1^\infty \frac{1}{x^2}\,dx$
$= \lim_{t\to\infty}\left[-\dfrac{1}{x}\right]_1^t = \lim_{t\to\infty}\left(-\dfrac{1}{t} + 1\right) = 1$. Converges to 1.

Example 2: $\displaystyle\int_1^\infty \frac{1}{x}\,dx$
$= \lim_{t\to\infty}[\ln x]_1^t = \lim_{t\to\infty}\ln t = \infty$. **Diverges.**

Example 3: $\displaystyle\int_0^1 \frac{1}{\sqrt{x}}\,dx$
$= \lim_{\varepsilon\to 0^+}\left[2\sqrt{x}\right]_\varepsilon^1 = \lim_{\varepsilon\to 0^+}(2 - 2\sqrt{\varepsilon}) = 2$. Converges to 2.

Inverse Trigonometric and Hyperbolic Integrals

Standard results (must know):

Integral	Result
$\int \frac{1}{\sqrt{a^2 - x^2}}\,dx$	$\arcsin\frac{x}{a} + C$
$\int \frac{1}{a^2 + x^2}\,dx$	$\frac{1}{a}\arctan\frac{x}{a} + C$
$\int \frac{1}{\sqrt{x^2 + a^2}}\,dx$	$\operatorname{arcsinh}\frac{x}{a} + C = \ln\left(x + \sqrt{x^2 + a^2}\right) + C$
$\int \frac{1}{\sqrt{x^2 - a^2}}\,dx$	$\operatorname{arccosh}\frac{x}{a} + C = \ln\left(x + \sqrt{x^2 - a^2}\right) + C \quad (x > a)$
$\int \frac{1}{a^2 - x^2}\,dx$	$\frac{1}{2a}\ln\left\lvert\frac{a+x}{a-x}\right\rvert + C$

Example 1: $\int \frac{1}{\sqrt{9 - x^2}}\,dx = \arcsin\frac{x}{3} + C$

Example 2: $\int_0^2 \frac{1}{4 + x^2}\,dx = \left[\frac{1}{2}\arctan\frac{x}{2}\right]_0^2 = \frac{1}{2}\cdot\frac{\pi}{4} = \frac{\pi}{8}$

Example 3 (completing the square): $\int \frac{1}{\sqrt{x^2 + 4x + 5}}\,dx$

$x^2 + 4x + 5 = (x+2)^2 + 1$. Let $u = x + 2$:

$$\int \frac{1}{\sqrt{u^2 + 1}}\,du = \operatorname{arcsinh} u + C = \ln\left(x + 2 + \sqrt{x^2 + 4x + 5}\right) + C$$

Mean Value of a Function

The **mean value** of $f(x)$ on $[a, b]$ is:

$$\bar{f} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

Example: Find the mean value of $f(x) = x^2 + 1$ on $[0, 3]$.
$\bar{f} = \frac{1}{3}\int_0^3 (x^2+1)\,dx = \frac{1}{3}\left[\frac{x^3}{3} + x\right]_0^3 = \frac{1}{3}(9+3) = 4$

Example 2: Find the mean value of $\sin x$ on $[0, \pi]$.
$\bar{f} = \frac{1}{\pi}\int_0^\pi \sin x\,dx = \frac{1}{\pi}[-\cos x]_0^\pi = \frac{1}{\pi}(1+1) = \frac{2}{\pi}$

Further Reduction Formula Examples

Derive $I_n = \int_0^{\pi/2} \cos^n x\,dx$
Write $\cos^n x = \cos^{n-1} x \cdot \cos x$. IBP with $u = \cos^{n-1} x,\ dv = \cos x\,dx$:
$I_n = [\cos^{n-1} x \sin x]_0^{\pi/2} + (n-1)\int_0^{\pi/2} \cos^{n-2} x \sin^2 x\,dx$
$= 0 + (n-1)\int_0^{\pi/2} \cos^{n-2} x(1-\cos^2 x)\,dx = (n-1)(I_{n-2} - I_n)$
$\Rightarrow nI_n = (n-1)I_{n-2} \Rightarrow \boxed{I_n = \frac{n-1}{n} I_{n-2}}$
With $I_0 = \frac{\pi}{2},\ I_1 = 1$: e.g. $I_4 = \frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2} = \frac{3\pi}{16}$.

Derive $J_n = \int x^n e^x\,dx$
IBP: $u = x^n,\ dv = e^x\,dx$:
$J_n = x^n e^x - n\int x^{n-1} e^x\,dx = x^n e^x - nJ_{n-1}$
So $J_0 = e^x + C,\ J_1 = xe^x - e^x + C,\ J_2 = x^2 e^x - 2xe^x + 2e^x + C = e^x(x^2 - 2x + 2) + C$.

Parametric Differentiation and Applications

For a curve $x = f(t),\ y = g(t)$:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} \qquad \frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt}$$

Example: $x = t^2,\ y = t^3 - 3t$. Find $\dfrac{dy}{dx}$ and stationary points.
$\dfrac{dy}{dx} = \dfrac{3t^2 - 3}{2t}$. Stationary points where $3t^2 - 3 = 0 \Rightarrow t = \pm 1$.
$t = 1$: $(1, -2)$; $t = -1$: $(1, 2)$.

Example 2 — Area under parametric curve:
Area $= \displaystyle\int y\,dx = \int y\,\frac{dx}{dt}\,dt$. For astroid $x = a\cos^3 t,\ y = a\sin^3 t$:
Area $= 4\displaystyle\int_0^{\pi/2} a\sin^3 t \cdot 3a\cos^2 t(-\sin t)\cdot(-1)\,dt = 12a^2\int_0^{\pi/2}\sin^4 t\cos^2 t\,dt = \frac{3\pi a^2}{8}$

Standard Substitutions — Quick Reference

Choose a substitution to simplify the integrand:

Integrand contains	Try substitution	Useful identity
$\sqrt{a^2 - x^2}$	$x = a\sin\theta$	$1 - \sin^2\theta = \cos^2\theta$
$\sqrt{a^2 + x^2}$	$x = a\tan\theta$ or $x = a\sinh u$	$1 + \tan^2\theta = \sec^2\theta$
$\sqrt{x^2 - a^2}$	$x = a\sec\theta$ or $x = a\cosh u$	$\cosh^2 u - \sinh^2 u = 1$
e^{ax} with trig	Complex exponentials or cyclic IBP	—

Rational in $\sin\theta, \cos\theta$	$t = \tan(\theta/2)$	$\sin\theta = \dfrac{2t}{1+t^2},\ \cos\theta = \dfrac{1-t^2}{1+t^2}$

Example — $t = \tan(\theta/2)$ substitution: $\displaystyle\int \frac{1}{1+\cos\theta}\,d\theta$

$\cos\theta = \dfrac{1-t^2}{1+t^2},\ d\theta = \dfrac{2}{1+t^2}\,dt.$

Integrand: $\dfrac{1}{1+\frac{1-t^2}{1+t^2}} \cdot \dfrac{2}{1+t^2} = \dfrac{1+t^2}{2} \cdot \dfrac{2}{1+t^2} = 1.$

So $\displaystyle\int 1\,dt = t + C = \tan\frac{\theta}{2} + C.$

Arc Length — Cartesian and Polar

Cartesian arc length: $L = \displaystyle\int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$

Parametric arc length: $L = \displaystyle\int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$

Polar arc length: $L = \displaystyle\int_\alpha^\beta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\,d\theta$

Example 1 — Cartesian: $y = \dfrac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$.

$\dfrac{dy}{dx} = x^{1/2}$, so $1 + (y')^2 = 1 + x.$

$L = \displaystyle\int_0^3 \sqrt{1+x}\,dx = \left[\tfrac{2}{3}(1+x)^{3/2}\right]_0^3 = \tfrac{2}{3}(8-1) = \frac{14}{3}$

Example 2 — Parametric: Cycloid $x = a(t - \sin t),\ y = a(1-\cos t)$, one arch $0 \le t \le 2\pi$.

$\dot{x} = a(1-\cos t),\ \dot{y} = a\sin t.$

$\dot{x}^2 + \dot{y}^2 = a^2(1 - 2\cos t + \cos^2 t + \sin^2 t) = a^2 \cdot 2(1-\cos t) = 4a^2\sin^2\tfrac{t}{2}.$

$L = \displaystyle\int_0^{2\pi} 2a\left|\sin\tfrac{t}{2}\right|\,dt = 4a\int_0^\pi \sin u\,du \cdot 2 = 8a.$

Surface Area of Revolution — Examples

About x-axis (Cartesian): $S = 2\pi \int_a^b y\sqrt{1+(y')^2}\,dx$

About x-axis (Parametric): $S = 2\pi \int_{t_1}^{t_2} y\sqrt{\dot{x}^2+\dot{y}^2}\,dt$

Example: The arc of $y = \sqrt{x}$ from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the surface area.

$y' = \dfrac{1}{2\sqrt{x}}$, so $1 + (y')^2 = 1 + \dfrac{1}{4x} = \dfrac{4x+1}{4x}$.

$S = 2\pi \int_0^1 \sqrt{x} \cdot \dfrac{\sqrt{4x+1}}{2\sqrt{x}}\,dx = \pi \int_0^1 \sqrt{4x+1}\,dx$

$= \pi \left[\dfrac{(4x+1)^{3/2}}{6}\right]_0^1 = \dfrac{\pi}{6}(5^{3/2}-1) = \dfrac{\pi}{6}(5\sqrt{5}-1)$

Numerical Integration — Reminder

Trapezium Rule: $\int_a^b f(x)\,dx \approx \dfrac{h}{2}\left[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)\right]$, where $h = \dfrac{b-a}{n}$.

Error: proportional to h^2. Halving the step size quarters the error.

Exam-tip — Further Calculus AQA:

- Volume: $\pi \int y^2\,dx$ not $\pi \int y\,dx$. The π is outside the integral.
- Surface area: has $2\pi y$ (circumference of strip), plus the arc-length element — distinguish from arc length itself.
- Substitution integrals: change limits when using definite integrals, OR change back to x before substituting limits.
- Reduction formulae in AQA: usually derived on the paper — show IBP clearly, collect I_n terms.
- Improper integrals: never write $\infty - \infty$; always take the limit first, then subtract.

- $\int \sec^2 x\,dx = \tan x + C$ and $\int \operatorname{sech}^2 x\,dx = \tanh x + C$ are both standard.

Complete Standard Integrals — AQA Further Calculus

Integral	Result
$\int \sec^2 ax\,dx$	$\dfrac{\tan ax}{a} + C$
$\int \sec x \tan x\,dx$	$\sec x + C$
$\int \csc^2 x\,dx$	$-\cot x + C$
$\int \cosh ax\,dx$	$\dfrac{\sinh ax}{a} + C$
$\int \sinh ax\,dx$	$\dfrac{\cosh ax}{a} + C$
$\int \operatorname{sech}^2 x\,dx$	$\tanh x + C$
$\int \dfrac{1}{\sqrt{a^2 - x^2}}\,dx$	$\arcsin \dfrac{x}{a} + C$
$\int \dfrac{1}{a^2 + x^2}\,dx$	$\dfrac{1}{a} \arctan \dfrac{x}{a} + C$
$\int \dfrac{1}{\sqrt{x^2 + a^2}}\,dx$	$\ln\left\lvert x + \sqrt{x^2 + a^2}\right\rvert + C$
$\int \dfrac{1}{\sqrt{x^2 - a^2}}\,dx$	$\ln\left\lvert x + \sqrt{x^2 - a^2}\right\rvert + C \quad (x > a)$

Maclaurin Series — Revision and Applications to Integration

Key Maclaurin expansions valid for all x unless stated:

Function	Expansion	Valid
e^x	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$	all x
$\sin x$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$	all x
$\cos x$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$	all x
$\ln(1+x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$	$-1 < x \leq 1$
$(1+x)^n$	$1 + nx + \frac{n(n-1)}{2!}x^2 + \cdots$	$\lvert x \rvert < 1$ (non-integer n)
$\arctan x$	$x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$	$\lvert x \rvert \leq 1$

Example — Approximate an integral using series:

Estimate $\int_0^{0.5} \frac{\sin x}{x}\,dx$ to 4 d.p.

$$\frac{\sin x}{x} = 1 - \frac{x^2}{6} + \frac{x^4}{120} - \cdots$$

$$\int_0^{0.5} \left(1 - \frac{x^2}{6} + \frac{x^4}{120}\right) dx = \left[x - \frac{x^3}{18} + \frac{x^5}{600}\right]_0^{0.5}$$

$$= 0.5 - \frac{0.125}{18} + \frac{0.03125}{600} \approx 0.5 - 0.00694 + 0.000052 \approx 0.4931$$

Example — Series for a composite function: Find the first 3 non-zero terms of $e^{\sin x}$.

$\sin x = x - \frac{x^3}{6} + \cdots$, so $e^{\sin x} = e^{x - x^3/6 + \cdots}$.

$$= 1 + \left(x - \frac{x^3}{6}\right) + \frac{1}{2}\left(x - \frac{x^3}{6}\right)^2 + \frac{1}{6}x^3 + \cdots$$

$$= 1 + x + \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^3}{6} + \cdots = 1 + x + \frac{x^2}{2} + 0 \cdot x^3 + \cdots$$

Partial Fractions and Integration — Combined Examples

Example 1: $\displaystyle\int \frac{3x+5}{(x+1)(x+2)}\,dx$

Partial fractions: $\displaystyle\frac{3x+5}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$.

$A = 2,\ B = 1$. So $\displaystyle\int \left(\frac{2}{x+1} + \frac{1}{x+2}\right) dx = 2\ln|x+1| + \ln|x+2| + C$.

Example 2 (repeated root): $\displaystyle\int \frac{x}{(x-1)^2(x+2)}\,dx$

Write $\displaystyle\frac{x}{(x-1)^2(x+2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2}$.

Solving: $A = \frac{2}{3}, B = \frac{1}{3}, C = -\frac{2}{3}$.

$\displaystyle\int = \frac{2}{3}\ln|x-1| - \frac{1}{3(x-1)} - \frac{2}{3}\ln|x+2| + C$

Integration of Trigonometric Powers

Use identities to reduce powers:

- $\cos^2 x = \dfrac{1+\cos 2x}{2}, \quad \sin^2 x = \dfrac{1-\cos 2x}{2}$
- $\sin x \cos x = \dfrac{\sin 2x}{2}$
- $\tan^2 x = \sec^2 x - 1$
- For odd powers of $\sin$ or $\cos$: peel off one factor and substitute.

Example 1: $\displaystyle\int \cos^4 x\,dx$

$$\cos^4 x = \left(\frac{1+\cos 2x}{2}\right)^2 = \frac{1 + 2\cos 2x + \cos^2 2x}{4} = \frac{1 + 2\cos 2x + \frac{1+\cos 4x}{2}}{4}$$

$$= \frac{3}{8} + \frac{\cos 2x}{2} + \frac{\cos 4x}{8}$$

$$\int \cos^4 x\, dx = \frac{3x}{8} + \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$$

Example 2: $\int \sin^3 x\, dx$

Write $\sin^3 x = \sin^2 x \cdot \sin x = (1 - \cos^2 x)\sin x$. Let $u = \cos x,\ du = -\sin x\, dx$:

$$\int (1 - u^2)(-du) = -u + \frac{u^3}{3} + C = -\cos x + \frac{\cos^3 x}{3} + C$$

Example 3: $\int \tan^3 x\, dx$

$$\tan^3 x = \tan x(\sec^2 x - 1) = \tan x \sec^2 x - \tan x$$

$$\int \tan^3 x\, dx = \frac{\tan^2 x}{2} - \ln|\sec x| + C$$

Wallis's Formula and Applications

From the reduction formula $I_n = \dfrac{n-1}{n} I_{n-2}$:

$$\int_0^{\pi/2} \sin^n x\, dx = \int_0^{\pi/2} \cos^n x\, dx = \begin{cases} \dfrac{(n-1)!!}{n!!} \cdot \dfrac{\pi}{2} & n \text{ even} \\ \dfrac{(n-1)!!}{n!!} & n \text{ odd} \end{cases}$$

where $n!! = n(n-2)(n-4)\cdots$. So $\int_0^{\pi/2} \sin^6 x\, dx = \dfrac{5 \cdot 3 \cdot 1}{6 \cdot 4 \cdot 2} \cdot \dfrac{\pi}{2} = \dfrac{5\pi}{32}$.

Wallis's product for π: Setting $n = 2m$ and comparing even and odd reduction chains:
$\dfrac{\pi}{2} = \dfrac{2 \cdot 2 \cdot 4 \cdot 4 \cdot 6 \cdot 6 \cdots}{1 \cdot 3 \cdot 3 \cdot 5 \cdot 5 \cdot 7 \cdots}$ (Wallis's product).
This arises directly from comparing I_{2n} and I_{2n+1} as $n \to \infty$.

Exam note — using Wallis: You will not be expected to quote Wallis's formula, but the reduction formula is examinable. Show each IBP step and collect terms carefully. The formula $I_n = \dfrac{n-1}{n} I_{n-2}$ with correct base cases $I_0 = \dfrac{\pi}{2},\ I_1 = 1$ is sufficient to evaluate all definite integrals of this type.

Quick Differentiation Reminders

Derivatives that arise frequently in Further Calculus questions:

- $\dfrac{d}{dx}(\arcsin x) = \dfrac{1}{\sqrt{1-x^2}}$
- $\dfrac{d}{dx}(\arctan x) = \dfrac{1}{1+x^2}$
- $\dfrac{d}{dx}(\operatorname{arcsinh} x) = \dfrac{1}{\sqrt{1+x^2}}$
- $\dfrac{d}{dx}(\operatorname{arccosh} x) = \dfrac{1}{\sqrt{x^2-1}} \quad (x > 1)$
- $\dfrac{d}{dx}(\sec x) = \sec x \tan x$